Mastering Diffusion

Week 4 - Denoising Diffusion Probabilistic Model

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Review



Variational Inference

$$\log p_\theta(x) = \log \int_z p_\theta(x,z)dz$$

Marginalization over latent variable z

$$= \log \int_z p_\theta(x,z)\frac{q_\phi(z|x)}{q_\phi(z|x)}dz$$

$$= \log \mathbb{E}_{z\sim q_\phi(z|x)}\left[\frac{p_\theta(x,z)}{q_\phi(z|x)}\right]$$

For tractability, sample from q, not the entire z

$$\geq \mathbb{E}_{z\sim q_\phi}\left[\log \frac{p_\theta(x,z)}{q_\phi(z|x)}\right]$$

For tractability, use Jensen's inequality

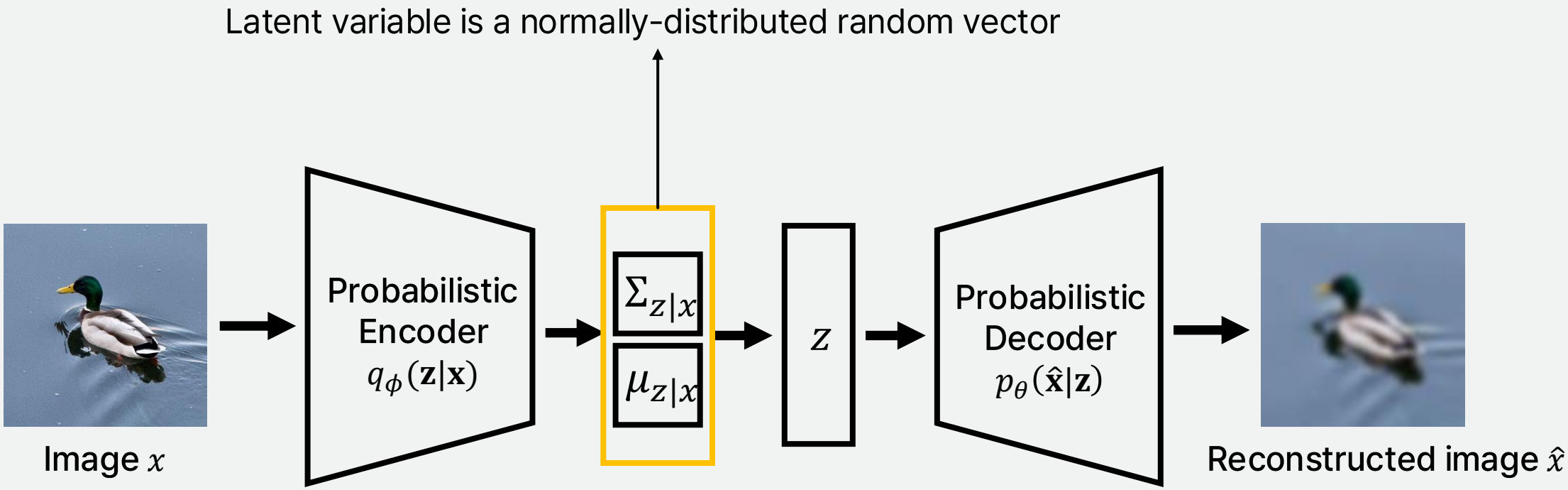
$$= \mathbb{E}_{z\sim q_\phi}[\log p_\theta(x|z)p(z)] - \mathbb{E}_{z\sim q_\phi}[q_\phi(z|x)]$$

$$= \mathbb{E}_{z\sim q_\phi}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$$

Variational bound from the last session

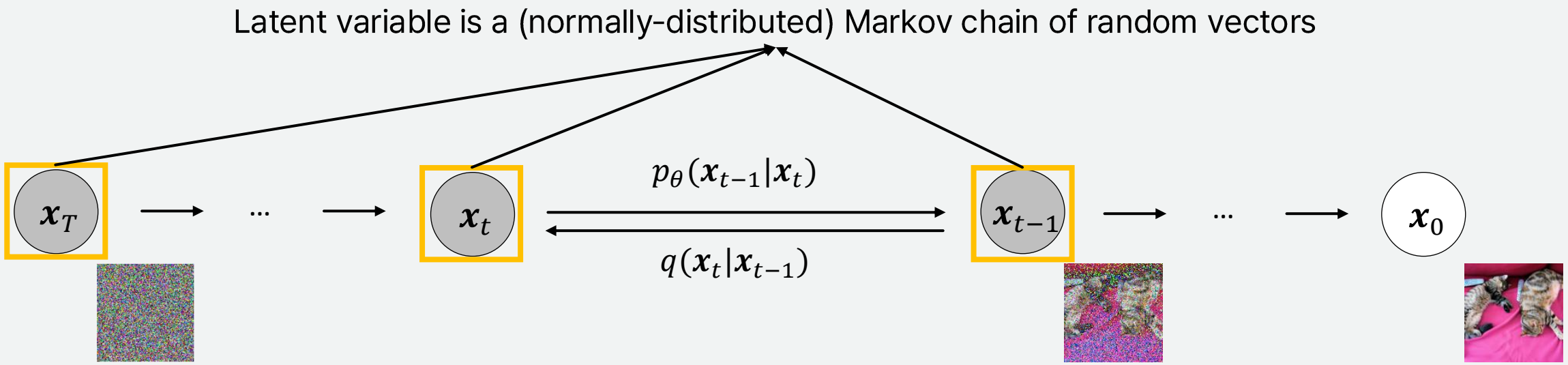


Latent variable in VAE is a random vector



$$\log p_\theta(x) = \log \int_z p_\theta(x, z) dz$$

Latent variable in DDPM is a Markov chain of random vectors



$$\log p_\theta(x_0) = \log \int_{x_{1:T}} p_\theta(x_0, x_{1:T}) dx_{1:T}$$

$$= \log \int_{x_{1:T}} p_\theta(x_{0:T}) dx_{1:T}$$

Image credit: https://huggingface.co/blog/annotated-diffusion

Notation in DDPM

- Probability density

$$p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$
$$\Leftrightarrow p(\boldsymbol{x}) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{||\boldsymbol{x} - \boldsymbol{\mu}||^2}{2\sigma^2}\right)$$

- Reparametrization trick (isolates the randomness into epsilon)

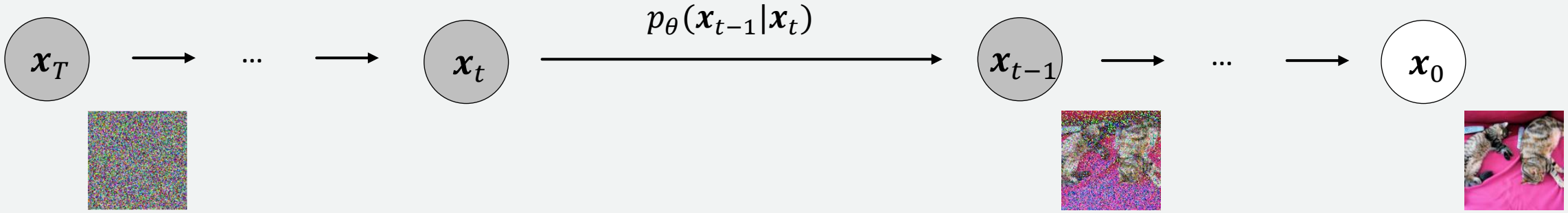
$$\boldsymbol{x} = \boldsymbol{\mu} + \sigma\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$
$$\Leftrightarrow \boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$$



Reverse process and forward process



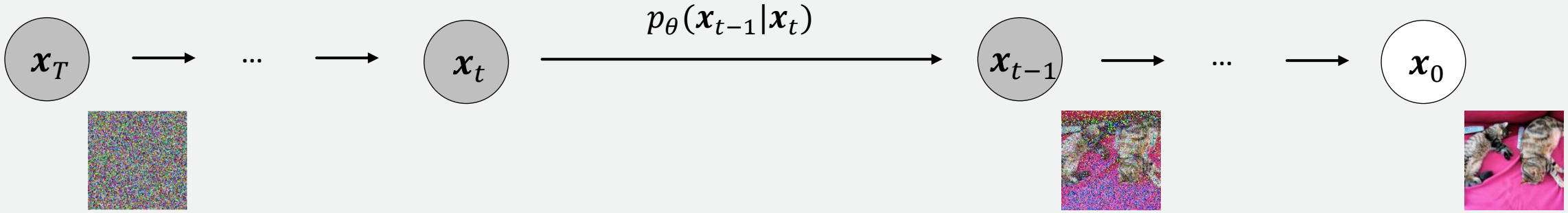
Reverse process (denoising)



Denoising across T timesteps

Image credit: https://huggingface.co/blog/annotated-diffusion

Reverse process (denoising)

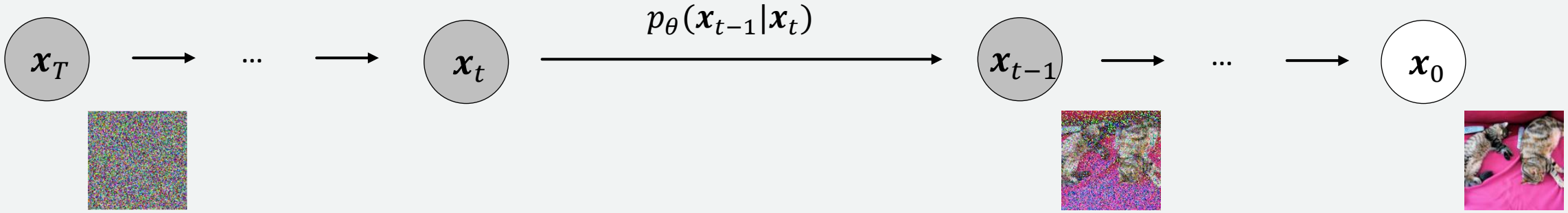


1) Markov chain

$$p_\theta(\boldsymbol{x}_{0:T}) \coloneqq p(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$$

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Reverse process (denoising)



1) Markov chain

$$p_\theta(\boldsymbol{x}_{0:T}) \coloneqq p(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$$

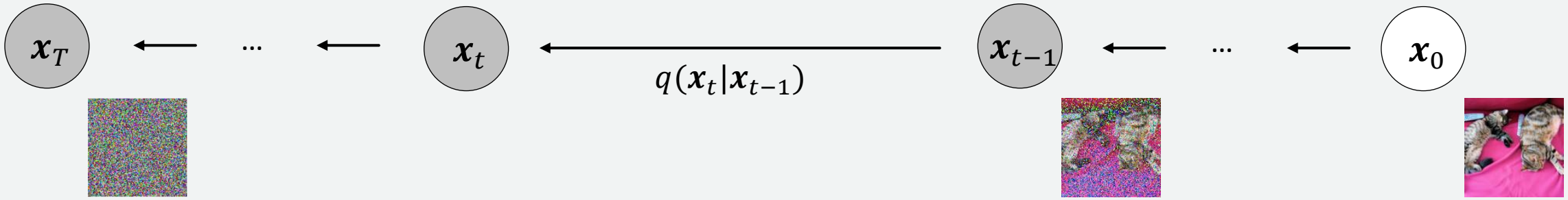
Pure noise Learned transition

2) Gaussian random vectors

$$p(\boldsymbol{x}_T) \coloneqq \mathcal{N}(\boldsymbol{x}_T; \mathbf{0}, \mathbf{I}) \qquad p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$$

Image credit: https://huggingface.co/blog/annotated-diffusion

Forward process (noising)



1) Markov chain

$$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) \coloneqq \prod_{t=1}^{T} q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$$

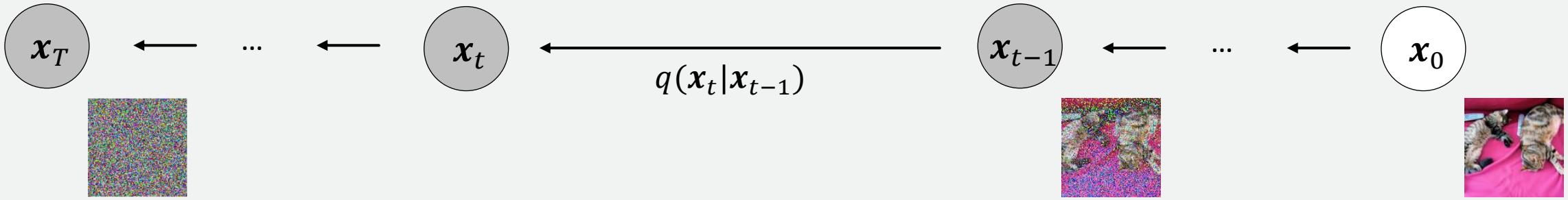
2) Gaussian random vectors

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I})$$

Variance schedule $\beta_1, \cdots, \beta_T$ controls adding noise

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Forward process is NOT trained



1) Markov chain

$$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) \coloneqq \prod_{t=1}^{T} q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$$

2) Gaussian random vectors

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I})$$

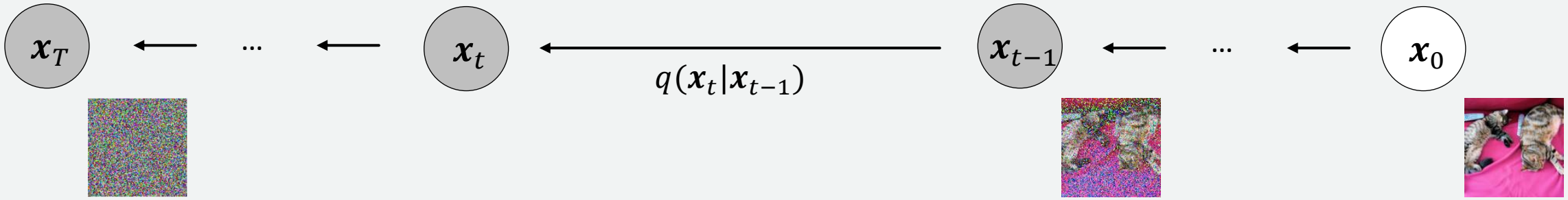
Variance schedule $\beta_1, \cdots, \beta_T$ controls adding noise

And variance schedule is **fixed (not learned)**

As a result, there is NO trainable component in the forward proce[illegible] q

Image credit: https://huggingface.co/blog/annotated-diffusion

Forward process (noising)



1) Markov chain

$$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) := \prod_{t=1}^{T} q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$$

2) Gaussian random vectors

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) := \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I})$$

$$\boldsymbol{x}_t = \sqrt{1-\beta_t}\boldsymbol{x}_{t-1} + \sqrt{\beta_t}\mathbf{I}$$
$$\mathrm{Var}(\boldsymbol{x}_t) = (1-\beta_t)\mathrm{Var}(\boldsymbol{x}_{t-1}) + \beta_t \mathbf{I}$$

Variance preserving: If $\mathrm{Var}(\boldsymbol{x}_{t-1}) = \mathbf{I}$, $\mathrm{Var}(\boldsymbol{x}_t) = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{I} = \mathbf{I}$

Image credit: https://huggingface.co/blog/annotated-diffusion

DDPM admits sampling at an arbitrary timestep

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \mathbf{I})$$

Definition of forward process

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\big(\boldsymbol{x}_t; \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\big), \qquad \alpha_t \coloneqq 1-\beta_t, \overline{\alpha_t} \coloneqq \Pi_{s=1}^{t}\alpha_s$$

Derivation from definition

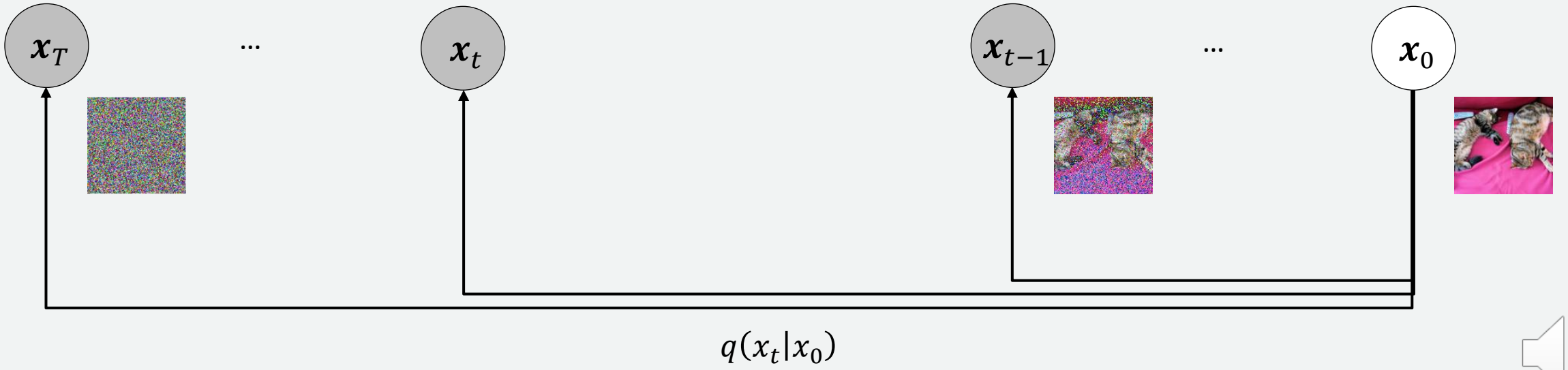


Image credit: https://huggingface.co/blog/annotated-diffusion

Given $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t\mathbf{I}\right)$,

show that $q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\right)$, $\quad \alpha_t \coloneqq 1-\beta_t$, $\quad \overline{\alpha_t} \coloneqq \Pi_{s=1}^{t}\alpha_s$

$\boldsymbol{x}_t = \sqrt{1-\beta_t}\boldsymbol{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_{t-1}$, $\quad \boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Let $\alpha_t \coloneqq 1-\beta_t$, then $\boldsymbol{x}_t = \sqrt{\alpha_t}\boldsymbol{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$

$x_t = \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}}\boldsymbol{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2}\right) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$

$= \sqrt{\alpha_t\alpha_{t-1}}\boldsymbol{x}_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-2} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$

Remember the Markov property, $\boldsymbol{\epsilon}_{t-2} \perp \boldsymbol{\epsilon}_{t-1}$

Let $\widetilde{\boldsymbol{\epsilon}}_t = \sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-2} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$ then $\widetilde{\boldsymbol{\epsilon}}_t \sim \mathcal{N}(0, \alpha_t(1-\alpha_{t-1}) + 1 - \alpha_t) = \mathcal{N}(0, 1-\alpha_t\alpha_{t-1})$

Now recursively, $\boldsymbol{x}_t = \sqrt{\alpha_t\alpha_{t-1}\cdots\alpha_1}\boldsymbol{x}_0 + \sqrt{1-\alpha_t\alpha_{t-1}\alpha_1}\boldsymbol{\epsilon}$, $\quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$

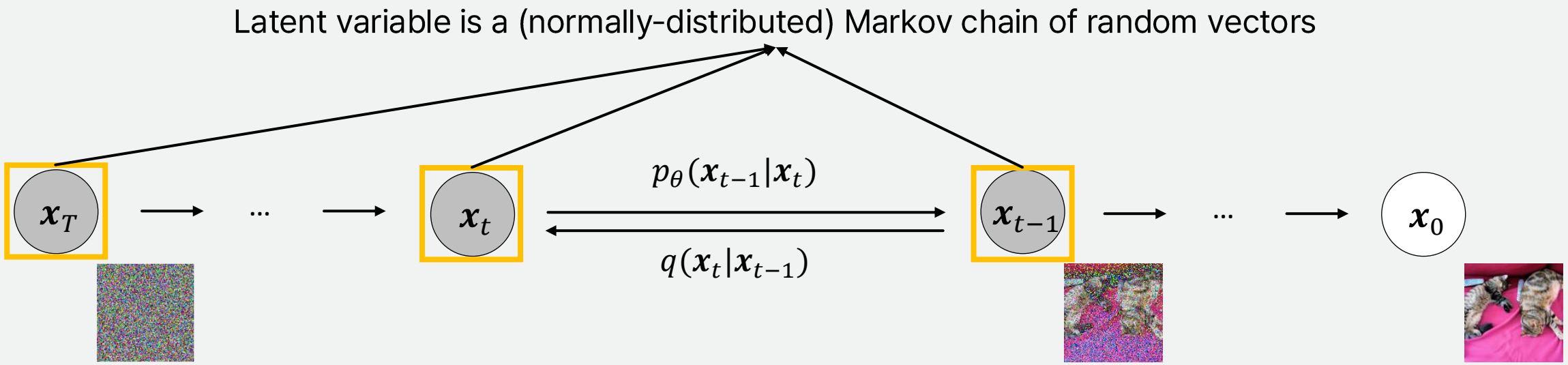
$\therefore\ q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\right)$



Training and sampling



Recap: Latent variable in DDPM is a Markov chain of random vectors



$$\log p_\theta(x_0) = \log \int_{x_{1:T}} p_\theta(x_0, x_{1:T}) dx_{1:T}$$

$$= \log \int_{x_{1:T}} p_\theta(x_{0:T}) dx_{1:T}$$

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Variational inference in DDPM (just the same)

$$-\log p_\theta(\boldsymbol{x}_0) = -\log \int_{\boldsymbol{x}_{1:T}} p_\theta(\boldsymbol{x}_{0:T}) d\boldsymbol{x}_{1:T}$$

$$= -\log \int_{\boldsymbol{x}_{1:T}} p_\theta(\boldsymbol{x}_{0:T}) \frac{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)} dx_{1:T}$$

$$= -\log \mathbb{E}_{\boldsymbol{x}_{1:T} \sim q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)} \left[\frac{p_\theta(\boldsymbol{x}_{0:T})}{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)} \right]$$

$$\leq \mathbb{E}_q[-\log \frac{p_\theta(\boldsymbol{x}_{0:T})}{q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0)}]$$

$$= \mathbb{E}_q[-\log p(\boldsymbol{x}_T) - \sum_{t=1}^{T} \log \frac{p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)}{q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})}]$$

No trainable component in the encoder q (data → latent)

Remember the Markov property

Loss function from variational inference

$$\mathbb{E}_q \left[D_{KL}\big(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \,||\, p(\boldsymbol{x}_T)\big) + \sum_{t=2}^{T} D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) - \log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1) \right]$$

Constant as we fix $p(\boldsymbol{x}_T) \coloneqq \mathcal{N}(\boldsymbol{x}_T; \mathbf{0}, \boldsymbol{I})$

KL divergence between the forward posterior and the model

Later incorporated into the left term

Loss function from variational inference

$$\mathbb{E}_q \left[D_{KL}\big(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \,||\, p(\boldsymbol{x}_T)\big) + \sum_{t=2}^{T} \underbrace{D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big)}_{\text{KL divergence between the } \textbf{forward posterior} \text{ and the model}} - \log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1) \right]$$

Bayes' rule tells us the reverse probability("forward posterior"), even though q is defined as a forward process

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t, x_0|x_{t-1})q(x_{t-1}, x_0)}{q(x_t, x_0)}$$
$$= \frac{q(x_t|x_{t-1})q(x_{t-1}, x_0)}{q(x_t, x_0)}$$
$$= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

Show $\mathbb{E}_q[-\log p(x_T) - \sum_{t=1}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}] = \mathbb{E}_q\left[D_{KL}(q(x_T|x_0) \,||\, p(x_T)) + \sum_{t=2}^{T} D_{KL}(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1)\right]$

$$\mathbb{E}_q[-\log p(x_T) - \textstyle\sum_{t=1}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}]$$

$$= \mathbb{E}_q[-\log p(x_T) - \textstyle\sum_{t=2}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})} - \log \frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}]$$

$$= \mathbb{E}_q[-\log p(x_T) - \textstyle\sum_{t=2}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} \cdot \frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} - \log \frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}]$$

$$= \mathbb{E}_q[-\log p(x_T) - \Sigma_{t=2}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} - \log \frac{q(x_1|x_0)}{q(x_T|x_0)} - \log \frac{p_\theta(x_0|x_1)}{q(x_1|x_0)}]$$

$$= \mathbb{E}_q[-\log \frac{p(x_T)}{q(x_T|x_0)} - \Sigma_{t=2}^{T} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} - \log p_\theta(x_0|x_1)]$$

$$= \mathbb{E}_q\left[D_{KL}(q(x_T|x_0) \,||\, p(x_T)) + \Sigma_{t=2}^{T} D_{KL}(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1)\right]$$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t, x_0|x_{t-1})q(x_{t-1}, x_0)}{q(x_t, x_0)}$$
$$= \frac{q(x_t|x_{t-1})q(x_{t-1}, x_0)}{q(x_t, x_0)}$$
$$= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

Forward posterior

Loss function from variational inference

$$\mathbb{E}_q \Big[\underbrace{D_{KL}\big(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \,||\, p(\boldsymbol{x}_T)\big)}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big)}_{L_{t-1}} - \underbrace{\log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)}_{L_0} \Big]$$

KL divergence between
the forward posterior and the model

1) **What is the posterior $q(\boldsymbol{x_{t-1}}|\boldsymbol{x_t}, \boldsymbol{x_0})$?**
2) What is the model $p_\theta(x_{t-1}|x_t)$?



Let's get posterior $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$. We already know the following three.

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I})$$

Definition of forward process

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\right), \qquad \alpha_t \coloneqq 1-\beta_t, \overline{\alpha_t} \coloneqq \Pi_{s=1}^{t}\alpha_s$$

Derivation from definition

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \frac{q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)}{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}$$

Bayes theorem + Markov property

Ignoring the normalization constant, $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \propto q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)$

Without showing, we use that the product of two isotropic Gaussian densities is still an isotropic Gaussian density.

$$\text{Let } q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \sim \mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t \mathbf{I}\right)$$



Density of isotropic Gaussian

$$p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$
$$\Leftrightarrow\ p(\boldsymbol{x}) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{||\boldsymbol{x}-\boldsymbol{\mu}||^2}{2\sigma^2}\right)$$

Let's get posterior $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) := \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I})$$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\right), \qquad \alpha_t := 1-\beta_t, \overline{\alpha_t} := \Pi_{s=1}^{t}\alpha_s$$

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \propto q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0)$$

Let $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t \mathbf{I})$

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \propto q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_0) \propto \exp\left(-\frac{1}{2\beta_t}||\boldsymbol{x}_t - \sqrt{\alpha_t}\boldsymbol{x}_{t-1}||^2\right) \times \exp\left(-\frac{1}{2(1-\bar{\alpha}_{t-1})}||\boldsymbol{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\boldsymbol{x}_0||^2\right)$$

$$= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)\boldsymbol{x}_{t-1}^2 - 2\left(\frac{\sqrt{\alpha_t}\boldsymbol{x}_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\boldsymbol{x}_0}{1-\bar{\alpha}_{t-1}}\right)\boldsymbol{x}_{t-1} + \cdots\right)\right)$$

■ ▲

$$\tilde{\beta}_t = \blacksquare^{-1} = \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)^{-1} = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\blacktriangle}{\blacksquare} = \left(\frac{\sqrt{\alpha_t}\boldsymbol{x}_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\boldsymbol{x}_0}{1-\bar{\alpha}_{t-1}}\right)\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\boldsymbol{x}_0$$

Now we have the closed-form solution of the posterior $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \sim \mathcal{N}\left(\tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t \mathbf{I}\right)$$

$$\tilde{\boldsymbol{\mu}}_t \coloneqq \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\boldsymbol{x}_0$$

$$\tilde{\beta}_t \coloneqq \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

1) Forward posterior is also an isotropic Gaussian

2) Its mean $\tilde{\boldsymbol{\mu}}_t$ is a **mixture of $\boldsymbol{x}_t$ and $\boldsymbol{x}_0$** according to the variance schedule



Loss function from variational inference

$$\mathbb{E}_q \Big[\underbrace{D_{KL}\big(q(\boldsymbol{x}_T|\boldsymbol{x}_0) \,||\, p(\boldsymbol{x}_T)\big)}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big)}_{L_{t-1}} - \underbrace{\log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)}_{L_0} \Big]$$

KL divergence between
the forward posterior and the model

1) What is the posterior $q(x_{t-1}|x_t, x_0)$?
2) **What is the model $\boldsymbol{p_\theta(x_{t-1}|x_t)}$?**



Recall that $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$ is defined as a Gaussian density.

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$$

In DDPM, we fix $\boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)$ as an untrained time-dependent constant $\sigma_t^2\mathbf{I}$.

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \sigma_t^2\mathbf{I})$$

Now we can simplify KL divergence between two isotropic Gaussians as a **L2 distance between their mean**.

$$\begin{aligned} L_{t-1} &\coloneqq D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) \\ &= \mathbb{E}_q\Big[\frac{1}{2\sigma_t^2}\,||\, \underbrace{\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0)}_{\text{forward posterior}} - \underbrace{\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)}_{\text{model}}||^2\Big] + C \end{aligned}$$

Motivation for reparameterization

$$L_{t-1} \coloneqq D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}||\,\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)||^2\right] + C$$

1) Directly approximating the mean is possible

$$\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) \coloneqq \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\boldsymbol{x}_0$$

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) \coloneqq ??$$

Motivation for reparameterization

$$L_{t-1} := D_{KL}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) \,||\, p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big) = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}||\,\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)||^2\right] + C$$

1) Directly approximating the mean is possible

$$\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) := \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\boldsymbol{x}_0$$

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) := ??$$

2) But we know the coeffcients and $\boldsymbol{x}_t$!
We only don't know $\boldsymbol{x}_0$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\big(\boldsymbol{x}_t;\, \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0, (1-\overline{\alpha_t})\mathbf{I}\big), \qquad \alpha_t := 1-\beta_t, \overline{\alpha_t} := \Pi_{s=1}^{t}\alpha_s$$

$$\boldsymbol{x}_t = \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\therefore \boldsymbol{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\boldsymbol{x}_t - \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t\right)$$

3) Actually we only don't know $\boldsymbol{\epsilon}_t$,
$\boldsymbol{\epsilon}_t$ is the **only** vector we need to model



$$\alpha_t \coloneqq 1 - \beta_t, \overline{\alpha_t} \coloneqq \Pi_{s=1}^{t} \alpha_s$$

Let's reparametrize $\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0)$ to $\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t)$

$$\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) \coloneqq \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\left(\frac{1}{\sqrt{\bar{\alpha}_t}}\left(\boldsymbol{x}_t - \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t\right)\right)$$

$$= \frac{\sqrt{\alpha_t\bar{\alpha}_t}(1-\bar{\alpha}_{t-1}) + \sqrt{\bar{\alpha}_{t-1}}\beta_t}{(1-\bar{\alpha}_t)\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t$$

$$= \frac{\alpha_t\sqrt{\bar{\alpha}_{t-1}}(1-\bar{\alpha}_{t-1}) + \sqrt{\bar{\alpha}_{t-1}}\beta_t}{(1-\bar{\alpha}_t)\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t$$

$$= \frac{\sqrt{\bar{\alpha}_{t-1}}(\alpha_t + \beta_t - \bar{\alpha}_t)}{(1-\bar{\alpha}_t)\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t$$

$$= \frac{1}{\sqrt{\alpha_t}}\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_t \qquad = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_t\right) \coloneqq \tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t)$$

We know — We do not know



The model only needs to learn the epsilon, the noise at timestep t.

$$\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_t\right)$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right)$$

Plug in reparametrized $\widetilde{\boldsymbol{\mu}_t}(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t)$ into L_{t-1}

$$L_{t-1} \coloneqq D_{KL}\big(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)\big) = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2} || \,\widetilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)||^2\right] + C$$

$$\widetilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_t\right) \qquad \boldsymbol{x}_t = \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$L_{t-1} - C = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2} || \,\widetilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)||^2\right]$$

$$= \mathbb{E}_q\left[\frac{1}{2\sigma_t^2} || \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_t\right) - \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right)||^2\right]$$

$$= \mathbb{E}_q\left[\frac{1}{2\sigma_t^2} || \frac{\beta_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\left(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right)||^2\right]$$

$$= \mathbb{E}_q\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)} || \left(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right)||^2\right]$$



Sampling standard normal noise is equivalent to sampling latent

$$L_{t-1} \coloneqq D_{KL}\big(q(x_{t-1}|x_t, x_0) \,||\, p_\theta(x_{t-1}|x_t)\big) = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2} || \,\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{x}_0) - \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)||^2\right] + C$$

$$\tilde{\boldsymbol{\mu}}_t(\boldsymbol{x}_t, \boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_t\right) \qquad \boldsymbol{x}_t = \sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

What was q?
Recall $x_{1:T} \sim q(x_{1:T}|x_0)$

$$\begin{aligned}
&L_{t-1} - C \\
&= \mathbb{E}_q\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)} || \,(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t))||^2\right] \\
&= \mathbb{E}_q\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)} || \,(\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}_t, t))||^2\right] \\
&= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\,\mathbf{t}}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)} || \,(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}, t))||^2\right]
\end{aligned}$$

Also take an expectation over data $\boldsymbol{x}_0$

$$\mathbb{E}_{\boldsymbol{x}_0,\,\boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\,\mathbf{t}}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)} || \,(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0 + \sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon}, t))||^2\right]$$

DDPM uses a simpler variant loss

$$\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\, \mathbf{t}}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\left|\left|\left(\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta\left(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0+\sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon},t\right)\right)\right|\right|^2\right]$$

$$L_{\text{simple}}=\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\, \mathbf{t}}\left[\left|\left|\left(\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta\left(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0+\sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon},t\right)\right)\right|\right|^2\right]$$

"Beneficial to sample quality (and simpler to implement)"

DDPM uses a simpler variant loss

$$\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\, \mathbf{t}}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\left|\left|\left(\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta\left(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0+\sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon},t\right)\right)\right|\right|^2\right]$$

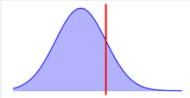
$$L_{\text{simple}}=\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\, \mathbf{t}}\left[\left|\left|\left(\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta\left(\sqrt{\overline{\alpha_t}}\boldsymbol{x}_0+\sqrt{(1-\overline{\alpha_t})}\boldsymbol{\epsilon},t\right)\right)\right|\right|^2\right]$$ *"Beneficial to sample quality (and simpler to implement)"*

Revisiting L_0

$L_{t-1}=\mathbb{E}[D_{KL}(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t,\boldsymbol{x}_0)\,||\,p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t))]$ KL divergence between distribution



$L_0=\mathbb{E}[-\log p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)]$ $\boldsymbol{x}_0$ is a fixed data (no stochasticity), while $p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)$ is a distribution. Let's compare the data with the mean of $p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1)$!



$$\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}\left[\left|\left|\left(\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta\left(\sqrt{\overline{\alpha_1}}\boldsymbol{x}_0+\sqrt{(1-\overline{\alpha_1})}\boldsymbol{\epsilon},1\right)\right)\right|\right|^2\right]$$
$$=\mathrm{C}\,\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}[||\left(\tilde{\boldsymbol{\mu}}_1(\boldsymbol{x}_1,\boldsymbol{x}_0)-\boldsymbol{\mu}_\theta(\boldsymbol{x}_1,1)\right)||^2]$$
$$\approx\mathrm{C}\,\mathbb{E}_{\boldsymbol{x}_0,\, \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}[||\underbrace{\boldsymbol{x}_0}-\underbrace{\boldsymbol{\mu}_\theta(\boldsymbol{x}_1,1)}||^2]$$

data mean of $\mathrm{p}_\theta(\mathrm{x}_0|\mathrm{x}_1)$

Simple loss is the L2 distance between the data and mean, ignoring the constants

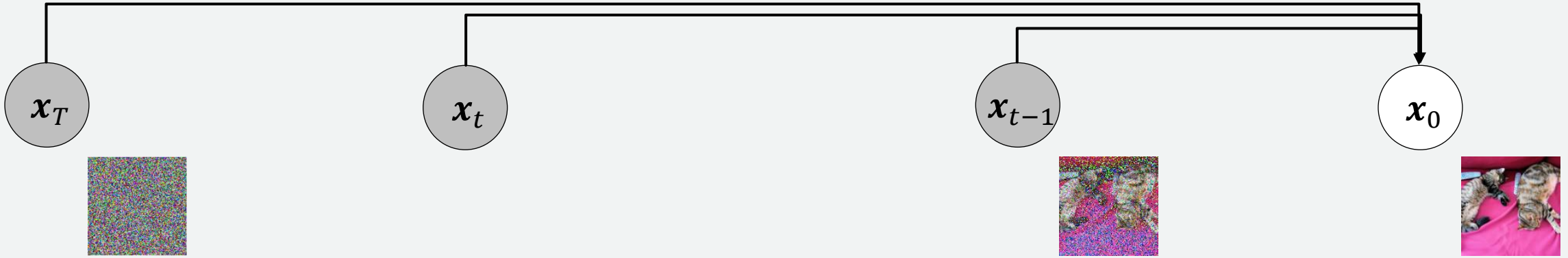
Training DDPM

$$\mathbb{E}_{x_0,\, \epsilon\sim\mathcal{N}(0,I),\, t}\left[\left|\left|\left(\epsilon - \epsilon_\theta\left(\sqrt{\overline{\alpha_t}}x_0 + \sqrt{(1-\overline{\alpha_t})}\epsilon, t\right)\right)\right|\right|^2\right]$$

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

DDPM learns to denoise from any timestep



Sampling from DDPM

Algorithm 2 Sampling

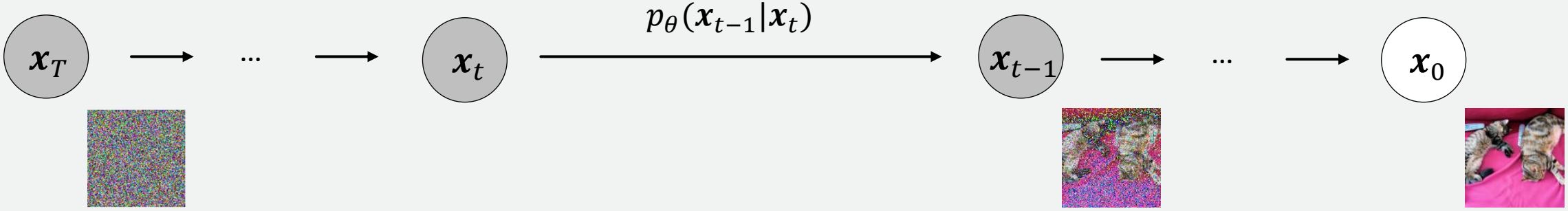
1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$$

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}} \boldsymbol{\epsilon}_\theta(x_t, t) \right)$$

$$\boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t) \coloneqq \sigma_t^2 \mathbf{I}$$

DDPM gradually denoises from the prior



Sampling from DDPM

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$$

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(x_t, t) \right)$$

$$\boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t) \coloneqq \sigma_t^2 \mathbf{I}$$

In principle, $\boldsymbol{x}_0$ can be generated from any timestep

$$\hat{\boldsymbol{x}}_0 = \frac{1}{\sqrt{\alpha_t}} (\boldsymbol{x}_t - \sqrt{1-\bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t))$$



Sampling from DDPM

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) := \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$$

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}} \boldsymbol{\epsilon}_\theta(x_t, t) \right)$$

$$\boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t) := \sigma_t^2 \mathbf{I}$$

In principle, x_0 can be generated from any timestep

This does NOT mean you can generate good samples from any timestep



Implementation of DDPM

In this part, we discuss the specific implementation of Ho et al., 2020, which **may not extend to other diffusion models**.

Discrete decoder

So far we discussed continuous densities, but image is NOT continuous

	0	3	
	8	15	24
		243	255

Re-scale

	-1		
	$-\frac{239}{255}$	$-\frac{225}{255}$	$-\frac{207}{255}$
		$\frac{231}{255}$	1

$x_0^i \in \{0, 1, 2, \cdots, 255\}$

$x_0^i \in \{-1, -\frac{253}{255}, -\frac{251}{255}, \cdots, 1\}$

Aligns with the standard normal prior

Discretize the modeled Gaussian density

$$p_\theta(\boldsymbol{x}_0|\boldsymbol{x}_1) = \mathcal{N}(\boldsymbol{x}_0; \boldsymbol{\mu}_\theta(\boldsymbol{x}_1, 1), \sigma_1^2 \mathbf{I})$$

(Pixel-level) Integrate Gaussian density over each bin

$$P(x_0^i = c) \coloneqq p_\theta\left(c - \frac{1}{255} \leq x_0^i \leq c + \frac{1}{255}\right)$$

$$= \int_{x_0^i - \frac{1}{255}}^{x_0^i + \frac{1}{255}} \mathcal{N}(x; \mu_\theta^i(\boldsymbol{x}_1, 1), \sigma_1^2)\, dx$$

(Image-level) Product of all pixels

$$P(\boldsymbol{x}_0 = \boldsymbol{c}) \coloneqq \prod_{i=1}^{D} P(x_0^i = c^i)$$

Variance schedule

Recall the forward process with variance schedule β_t

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I})$$

Variance schedule β_t is set increasingly linear, $\beta_1 = 10^{-4}, \beta_T = 0.02$.

This is small compared to the pixel range [-1,1].

Covariance of $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$

Recall that in DDPM, we fix $\boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)$ as an untrained time-dependent constant $\sigma_t^2\mathbf{I}$.

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \sigma_t^2\mathbf{I})$$

So what exactly is $\sigma_t^2\mathbf{I}$?

$\sigma_t^2 = \beta_t$ Variance of forward process $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$

$\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ Variance of forward posterior $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)$

→ Ho et al., 2020 reports similar results for above two

Upcoming topics

Week 5
Diffusion Sampling 2

SDE/score matching perspective
could be more intuitive to understand DDPM

Week 6
Various diffusion spaces

DDPM works on continuous space.
Diffusion models for **discrete space**(e.g., tokens)? For **latent space**?

Week 7
Technical components

DDPM deals with unconditional generation. What about **conditional generation**?

DDPM requires a lot of timesteps for sampling.
Can we **reduce the number of steps**?

Diffusion language model

Thank you!

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